# Chapter 2: Oregon Code And Compliance Map

This chapter identifies the source of every rule that controls the house. Oregon's system is a stack: state laws on top, agency rules under them, specialty codes under those, and the local building department running the day-to-day process. Once the layers are sorted, most code questions become lookups.

This chapter is the orientation to those sources. The Legal, Permits, Owner-Builder Rules, And Inspections chapter contains the legal detail.

## 2.1 The Oregon Rule Stack

Oregon does not use a single residential code. State law authorizes a family of specialty codes, administrative rules, and local procedures that together control building work.

> **Simple**
>
> Who makes the rules for your house? The rules come in layers. The legislature passes laws. State agencies write the detailed rules. Those rules adopt technical codes for building, electrical, plumbing, mechanical, and energy work. Your local building department runs the permits and inspections under all of it. The bottom layer is your approved drawings plus each product's installation instructions. Each layer contains the answer to a different kind of question.

> **Professional**
>
> The rule stack, top to bottom:

| Layer | What it does | What it controls for this house |
|---|---|---|
| Oregon Revised Statutes (ORS) | State laws passed by the legislature | Authority for building codes, contractor licensing, owner-builder rules, disclosures, and enforcement |
| Oregon Administrative Rules (OAR) | Agency rules that implement state law | Adoption of specialty codes, permit procedures, certificate-of-occupancy procedures, agency program detail |
| Oregon specialty codes | Technical building, electrical, plumbing, mechanical, and energy rules | The actual house systems and their inspections |
| Local jurisdiction process | Plan review, permit issuance, inspection scheduling, correction notices, approval records | The local submission and inspection workflow |

| | | |
|---|---|---|
| Approved plans and engineering | Project-specific drawings, calculations, specifications, and stamped sheets | The legal design for this house in particular |
| Referenced standards | Standards adopted by the codes (ASTM, ACI, ASCE, ASHRAE, NFPA, UL, ANSI, TMS, and others) | Test methods, materials, equipment ratings, fire ratings, listed assemblies |
| Manufacturer instructions | Required installation rules for listed products | How a specific product must be installed, supported, sealed, vented, tested, and maintained |

> The stack's structure stays the same across code cycles; the adopted editions change. The adopted edition of every code is confirmed with the permit-issuing jurisdiction before plan submittal.

> **Code**

```
The state building code exists by statute.
```

```
ORS chapter 455 authorizes the specialty codes and the Building Codes Division (BCD), sets
uniform permit, inspection, and certificate-of-occupancy requirements (ORS 455.055), and
includes the owner-built structural exemption (ORS 455.320) and the rural-county exemption
framework (ORS 455.330-455.350).
```

```
Source: ORS chapter 455.
```

```
Contractor licensing and the owner-builder role come from a second statute.
```

```
ORS chapter 701 creates the Construction Contractors Board (CCB), the owner-occupier
exemption from licensure with its intent-to-sell test (ORS 701.010), and the
homeowner-statement framework (ORS 701.325).
```

```
Source: ORS chapter 701.
```

```
BCD administers the codes by administrative rule.
```

```
OAR chapter 918 adopts each specialty code and sets the statewide administration; division
480 covers one- and two-family dwellings.
```

```
Source: OAR chapter 918.
```

```
The residential code in force is the 2023 edition.
```

```
2023 ORSC, based on the 2021 IRC; effective October 1, 2023, mandatory April 1, 2024. The next edition (2026 ORSC, 2024 IRC base) is expected October 1, 2026; confirm the edition in force at permit application.
```

```
Source: OAR 918-480-0005; BCD adopted-codes page, oregon.gov/bcd.
```

```
Local building departments run the state code, not their own.
```

```
Each city or county administers the state specialty codes through its building program; local amendments apply only where the state allows them.
```

```
Source: ORS 455.148-455.153; the local jurisdiction's adopted program.
```

## 2.2 The Main Code Lanes For This House

A concrete house still contains the same systems as any other dwelling: structure, water, drains, power, heat, ventilation, insulation, egress, fire safety, sanitation, and a final approval before anyone moves in. The clean way to read the code is one lane at a time.

> **Simple**
>
> Each part of the house has its own code. The building itself follows the residential code. The wiring follows the electrical code. The pipes follow the plumbing code. The heating and ventilation follow the mechanical code. Septic systems and wells fall under separate state agencies. The first step for any code question is choosing the matching lane from the table below.
>
> **Professional**

| Work area | Primary lane | What that lane controls |
|---|---|---|
| Residential building | Oregon Residential Specialty Code (ORSC) | Building planning, structure, foundations, walls, floors, roofs, stairs, openings, fire separation, energy (Chapter 11), inspections, occupancy |
| Electrical | Oregon Electrical Specialty Code (OESC) | Service, panels, grounding, bonding, branch circuits, receptacles, lighting, protection devices, electrical inspections |

| | | |
|---|---|---|
| Plumbing | Oregon Plumbing Specialty Code (OPSC) | Water supply, drain-waste-vent, fixtures, water heaters, backflow, tests, plumbing inspections |
| Mechanical | ORSC mechanical chapters first; Oregon Mechanical Specialty Code (OMSC) for what they do not cover | HVAC equipment, ventilation, exhaust, ducts, hydronics, appliance clearances, mechanical inspections |
| Fuel gas | ORSC Chapter 24 | Gas piping, appliance connections, and the gas pressure test |
| Onsite wastewater | DEQ rules, often delegated to county health | Septic feasibility, permits, design, installation, inspection, ongoing maintenance |
| Water wells | Oregon Water Resources Department (OWRD) | Well construction, well logs, water-source legality |
| Fire and fuel systems | State Fire Marshal rules and product listings | LP gas licensing, tanks, special fuel systems |
| Workplace safety | Oregon OSHA, once anyone is hired | Trenching, fall protection, equipment, jobsite safety duties |
| Engineering and standards | Approved engineering and adopted standards (ACI, ASCE, ASTM, ASHRAE, NFPA, UL, ANSI, TMS) | Concrete design, loads, reinforcing, materials, test methods, ratings, listed assemblies |

**Code**

Residential building work follows the ORSC.

2023 edition, based on the 2021 IRC, adopted by OAR 918-480-0005; covers detached one- and two-family dwellings and townhouses up to three stories, including the residential energy provisions in Chapter 11 and the fuel-gas provisions in Chapter 24.

*Source: OAR 918-480-0005; 2023 ORSC.*

Electrical work follows the OESC.

2023 edition: the 2023 NEC plus the Oregon amendments in Table 1-E (amended through January 1, 2026). The amendments matter; Oregon strikes or rewrites several national GFCI and AFCI rules.

*Source: OAR 918-305-0100; OESC Table 1-E, oregon.gov/bcd.*

Plumbing work follows the OPSC.

2023 edition, based on the 2021 UPC, effective February 9, 2024. Fuel-gas piping is not in the OPSC lane in Oregon; it lives in ORSC Chapter 24.

*Source: OAR 918-750-0110; 2023 OPSC.*

Mechanical work follows the ORSC first, then the OMSC.

The ORSC mechanical chapters govern dwellings directly; the 2025 OMSC (effective October 1, 2025, mandatory April 1, 2026) fills what they do not address.

*Source: OAR 918-440; 2025 OMSC; BCD mechanical program page.*

Septic systems follow DEQ's onsite wastewater rules.

Site evaluation, permit, installation, pre-cover inspection, certificate of satisfactory completion, and (from January 1, 2026) an ongoing operation-and-maintenance contract requirement.

*Source: OAR 340-071 (2026 rules); OAR 340-073.*

Wells follow OWRD's construction standards.

A landowner may drill on their own land under a Landowner Well Permit, meeting the same standards as licensed drillers; the well log goes to OWRD.

*Source: OAR 690-205-0175; OAR 690-200 through 690-240.*

LP gas work is licensed with no homeowner exemption.

Any LP gas installation, fitting, or piping work requires a State Fire Marshal licensed installer.

```
Source: ORS 480.432; OAR chapter 837 division 30.
```

```
Jobsite safety rules attach when anyone is hired.
```

```
Oregon OSHA construction rules (excavation, fall protection, equipment) apply to
employers; hiring the first paid worker creates the duty.
```

```
Source: ORS 654.005; OAR chapter 437 division 3.
```

This is a concrete-home book, so wood-framing rules are not reproduced. Wood appears only where this project uses it: roof framing if the roof is wood-framed, formwork, opening framing, fire separation, or a comparison against the concrete or masonry detail.

## 2.3 Which Document Answers Which Question

The table below matches common construction questions to the kind of answer each one needs.

| Question | First place to look |
|---|---|
| Am I legally allowed to act as the owner-builder? | ORS, OAR, BCD homeowner guidance, and the legal chapter of this book |
| Which permit path applies to this work? | The local building department or Oregon ePermitting |
| What must this wall, slab, wire, pipe, vent, drain, appliance, or window satisfy? | The applicable specialty code and the approved plans |
| Who must design this detail? | The code's prescriptive path, the engineer of record, or the building official |
| How is this product installed? | Its listing, label, and manufacturer instructions |
| How is this work tested? | The applicable code, referenced standard, or inspector requirement |
| Can I cover this work? | A passed inspection, plus your own photos and records |
| Can I occupy the home? | The certificate of occupancy (CO) or a lawful temporary CO |

A few failure patterns show up over and over when owner-builders meet the code system, so they are named here:

- A product brochure does not override code. Manufacturer instructions are required, but they cannot authorize an installation the code forbids.
- A construction video does not override the approved drawings.
- A habit from another state does not override Oregon. Codes differ by state; a practice that is correct elsewhere can be wrong here.
- A summary in this book does not override the official source. This book restates the code for daily use. Where the two differ, the adopted code text controls.
- A stronger-looking assembly is not automatically a compliant one. Compliance is judged on the whole assembly under the actual site conditions.

That last point applies most strongly to concrete. Adding concrete can improve a design, but it also adds load, changes seismic behavior, complicates reinforcement, and can interrupt insulation continuity or connection details. Added thickness does not establish compliance on its own; the complete assembly is verified under the actual site conditions.

## 2.4 Prescriptive, Engineered, And Alternative Paths

The code recognizes three design paths. This house uses at least two of them.

**Simple**

A prescriptive path is a fixed set of code instructions: the code states exactly what to build, and you build to those values. An engineered path is a custom design: a licensed engineer sizes the system for your actual house, soil, and loads. An alternative-method path is a special approval: the building official agrees that a different way still meets the code's intent.

**Applied**

Use this flow to classify any structural or envelope element:

1. Check whether an ORSC prescriptive section applies to the element within all its limits: geometry, seismic design category, load, height.
2. If yes, the prescriptive path is available. Confirm with the engineer of record that the prescriptive values fit the actual application.
3. If no, the element is designed under ORSC R301.1.3 (engineered design) using the relevant standard: ACI 318, ACI 332, PCA 100, TMS 402, ASCE 7.
4. If a non-prescriptive method is preferred (an innovative system, an unusual material, a tested assembly), request alternative-method approval from the building official under ORSC R104.11 before plan review.

5. Include the chosen path in the permit submittal: the prescriptive table cited, the engineered drawings stamped, or the approval letter on file.

**Professional**

| Path | Best fit for this project |
|---|---|
| Prescriptive | Ordinary details that fall clearly inside the code tables and standard assumptions |
| Engineered | Reinforced concrete, unusual foundations, retaining conditions, seismic design, large openings, complex load paths, nonstandard assemblies |
| Alternative method | Innovative systems, unusual materials, and tested assemblies that meet the code's intent by another route |

For most concrete and reinforced-masonry residences in western Oregon, engineered design is the default. Seismic Design Category (SDC, the code's classification of a site's earthquake demand) D0, D1, or D2 covers the Willamette Valley, the I-5 corridor, and the coast. Several prescriptive paths, notably ORSC R608 for above-grade concrete walls, exist only in SDC C, which maps to most of eastern Oregon. The Structural System And Concrete Design chapter states the specific triggers: seismic design category, two-wythe walls, irregular building conditions, nonstandard wall heights, expansive soil.

A concrete or masonry dwelling stays under the residential code by occupancy. Moving its above-grade walls to engineered design does not move the whole project to the commercial structural code. The dwelling remains a residential-code project for permitting, inspection, and occupancy; only its structural design method becomes nonprescriptive engineering, which references the same standards the commercial code uses (ACI 318, ACI 332, ASCE 7). A true jump into the commercial structural code would come from an occupancy or use change beyond the residential code's scope, not from seismic category alone.

The prescriptive sections in R608 (concrete walls) and R606 (masonry walls) remain useful as vocabulary and as minimum-expectation references. They are not a substitute for engineered design of a nonstandard concrete home.

An alternative-method request is an evidence package. The building official may accept written, certified inspection or evaluation reports from approved agencies or individuals as part of that evidence (ORSC R104.4). An out-of-state evaluation report (an ICC-ES report, from the International Code Council's Evaluation Service, or an IAPMO UES report, from IAPMO's Uniform Evaluation Service, or another jurisdiction's compliance finding) can be submitted this way. No rule obligates the building official to accept it; acceptance is that official's decision, case by case. An applicant relying on an outside report confirms its acceptance with the specific building official before plan review, since the code sets no statewide rule either requiring or forbidding such reports.

If the building official denies the request, the residential code requires a written response stating the reasons the alternative was not approved (ORSC R104.11). The code requires the reasons but does not dictate their form. A denial is most useful on appeal when it names which equivalency criterion (quality, strength, effectiveness, fire resistance, durability, or safety) the official found unmet, because the appeal is reviewed against those same criteria.

The standing practices for this project:

- Prescriptive details apply where they fit the geometry, soil, and seismic conditions.
- Engineered details apply where the system carries unusual demand.
- Any alternative-method approval is identified before plan review, not during an inspection.
- Product listings and installation instructions stay with the permit record.
- Field changes that touch approved work are treated as design changes.

Open code questions are resolved before concrete placement. Exposing and correcting a cured-in error requires demolition, and some errors cannot be corrected after the pour.

**Code**

```
Buildings must be designed for the loads in the code.
```

```
 Where a design exceeds the prescriptive limits, R301.1.3 sends it to accepted engineering
 practice, with stamped drawings and calculations from the engineer of record.
```

```
 Source: ORSC R301.1 and R301.1.3 (2023 edition).
```

```
Seismic design category sets which structural rules apply.
```

```
 SDC D0, D1, and D2 trigger expanded reinforcement for foundations, walls, and diaphragms.
```

```
 Source: ORSC R301.2.2 (2023 edition).
```

```
Above-grade concrete walls must have a recognized design basis.
```

```
 Exterior above-grade concrete walls must comply with PCA 100 or be designed under ACI 318,
 except that detached one- and two-family dwellings in SDC C may instead use the
 prescriptive insulating-concrete-form provisions (R608). In SDC D0-D2, which covers most
 of western Oregon, that exception does not apply, so the walls are engineered; ACI 332-20
 provides residential-scoped provisions within its limits.
```

*Source: ORSC R301.2.2.5 (2023 edition); PCA 100; ACI 318; ACI 332-20.*

A one- or two-family dwelling stays under the residential code by occupancy.

A detached one- or two-family dwelling is governed by the ORSC regardless of its structural material; seismic design category does not move it to the commercial structural code. Above-grade concrete or masonry walls outside SDC C are engineered under the referenced standards through nonprescriptive design, while permitting, inspection, and occupancy stay under the ORSC.

*Source: ORSC R301.1.3 and R301.2.2.5 (2023 edition).*

An irregular building condition removes the prescriptive path for the whole story.

Eight listed irregularity conditions (vertical offsets, large diaphragm openings, mass irregularities, framing above masonry or concrete, and others) each trigger engineered analysis.

*Source: ORSC R301.2.2.6 (2023 edition).*

Prescriptive concrete and masonry walls exist, with tight limits.

R608 (above-grade concrete) is available only in SDC C, which excludes most of western Oregon; R606 (masonry) limits bar size, spacing, lap, grout, mortar, and seismic category.

*Source: ORSC R608; ORSC R606; TMS 402-22.*

The building official may approve an alternative material, design, or method.

Approval requires that the alternative is satisfactory and complies with the intent of the code, and is at least equivalent in quality, strength, effectiveness, fire resistance, durability, and safety to what the code prescribes; the submittal includes supporting test data and calculations.

*Source: ORSC R104.11 (2023 edition).*

A denied alternative gets a written response with reasons.

> Where the building official does not approve the alternative, the official responds in writing stating the reasons the alternative was not approved.

> *Source: ORSC R104.11 (2023 edition).*

> The building official may accept certified reports from approved agencies.

> The official may accept written reports of inspection or evaluation from approved agencies or individuals; the reports are in writing and certified by a responsible officer of the agency or by the responsible individual. Approval of the agency is the official's discretionary act, so an out-of-state or third-party evaluation report is accepted only at the official's discretion.

> *Source: ORSC R104.4 (2023 edition).*

## 2.5 Statewide Alternate Methods

Beyond a project-specific approval, Oregon provides a state-level route that settles a design method for the whole state at once.

**Simple**

A statewide alternate method is a ruling from the state Building Codes Division that approves one design approach for use anywhere in Oregon. A local building official's approval under the alternative-materials rule applies to a single project in that one jurisdiction. A statewide alternate method applies in every Oregon jurisdiction: once the state issues it, each building department must allow the approved method within the limits the ruling sets. Using one is optional. Where a ruling applies to a design, an applicant can build to it without re-proving the method to each jurisdiction.

**Professional**

The Building Codes Division publishes the current list of statewide alternate methods, and the separate request process that produces new ones, on its website (oregon.gov/bcd). Each ruling is a numbered PDF. Each ruling is published in a fixed format:

- a header naming the ruling number and subject
- a statement of legal effect: building officials must approve the addressed method, use of the method is at the applicant's discretion, and the ruling does not limit a building official's authority to consider other alternate-method requests on the same subject
- a Code References list: the applicable specialty-code section and the consensus standards involved
- the issue date and any update date
- a Background section stating the problem the ruling addresses

- a Discussion section stating the engineering basis
- a Conclusion
- the operative Alternate Method Path: a numbered list of technical requirements at full code specificity

Statewide Alternate Method No. 15-01, Cross-Laminated Timber Seismic Force-Resisting Systems (cross-laminated timber, or CLT, is a structural panel made of glued lumber layers), issued January 15, 2015 and updated October 1, 2025, is four pages long. Its Code References list names the structural specialty-code section and five consensus standards (ASCE 7-22, ASCE 41-23, ANSI/APA PRG 320-2019, ANSI/AWC SDPWS-2021, ANSI/AWC NDS-2024). Its Alternate Method Path is seven numbered sections specifying seismic design factors, ductile yield modes, overstrength load combinations, structural modeling, drift limits, deformation compatibility, and limits on holes and notches, including a modified ASCE 7 table row. No current statewide alternate method addresses residential concrete or masonry wall design specifically; the cross-laminated-timber ruling is the format example, and its subject is the commercial structural code rather than the residential code.

**Code**

```
State-level rulings on acceptable materials, designs, or methods have statutory authority.
```

```
Any person, or a building official, may ask the Building Codes Division director to rule
on the acceptability of a material, design, or method under the state building code; a
non-official's request must be made and ruled on before the method is used.
```

```
Source: ORS 455.060.
```

```
A statewide alternate method applies in every jurisdiction within its scope.
```

```
All jurisdictions administering the state building code must enforce statewide code
interpretations and must allow the use of alternate-method rulings consistent with the
original scope of the ruling; a building official's only judgment is whether a specific
project falls inside that scope. Noncompliance can result in suspension or revocation of a
building official's, inspector's, or plans examiner's certification.
```

```
Source: OAR 918-008-0110.
```

## 2.6 Who Runs Your Building Department

The department that reviews the plans and inspects the work is not the same organization in every county. The arrangement sets which organization reviews and inspects the work; the code that applies is the same across all three.

**Simple**

Not every county runs its own building department. Oregon uses three arrangements. A county or city can run its own program. The state Building Codes Division can run a county's program directly with state staff. Or the state can hire a private firm to run it under contract. Across all three, the code, the inspections, and the building official's authority are the same. What differs is which organization staffs the department and which contact information you use.

**Professional**

Which arrangement applies is set per jurisdiction under state law. When a county does not take on its own building inspection program, the Building Codes Division director contracts with another municipality or a private firm, or staffs the program with state employees (ORS 455.148(6)). That single statute is the basis for both the state-direct model and the privately contracted model.

The department for a specific parcel is found through the Building Codes Division's Local Building Department Directory (oregon.gov/bcd). The Directory names the department that currently administers each permit service (structural, mechanical, electrical, plumbing) for a given city or county. It does not label the result as county-run, state-direct, or privately contracted, so the arrangement is confirmed by asking the department directly. Northwest Code Professionals, based in Eugene, is a confirmed private building-department contractor in Oregon. A candidate parcel's arrangement is verified once the site search names specific counties, and it is current only at lookup time.

Authority does not vary by arrangement. A private firm's inspector or plan reviewer must hold the same Oregon certification as a county-employed one, and holds the same authority to issue correction notices and to sign off on work; the certification attaches to the role of building official, inspector, or plans examiner, not to the employer (OAR 918-098-1470; ORS 455.080). A municipality treats a licensed firm's inspections and plan reviews as its own and bears no liability for the firm's work (ORS 455.148(9) and (10)); that liability rule does not reduce the firm's authority.

In a county where the Building Codes Division administers the program directly, the state-employed building official fills the role of the building official for that jurisdiction. That official holds the same alternative-materials approval authority under the residential code as a county-employed building official. An alternative-materials request does not route to Salem because the state administers the program; the assigned building official decides it, following the same alternative-materials process: a written approval, or a written denial stating the reasons the alternative was not approved. Where that official wants a second opinion on a novel proposal, they can request a Statewide Code Interpretation (a Building Codes Division ruling on how a code provision applies) under OAR 918-008-0085.

Stop-work authority follows the same pattern. The Oregon Residential Specialty Code does not adopt the model code's stop-work-order section (R114). Stop-work authority instead comes from the municipality's own ordinance, an authority the state building code statute expressly preserves (ORS 455.020(4)). The certified building official exercises it whether the department is county-run, state-direct, or a private contract.

**Code**

A county that does not assume its own program is served by the state or a contractor.

Where a county does not notify the director that it will administer and enforce a building inspection program, the director contracts with a municipality or other person, or uses state employees or agencies, to run the program.

*Source: ORS 455.148(6).*

A licensed firm's inspectors and plan reviewers hold the full authority of their certified role.

Certification and duties attach to the Oregon-certified role of building official, inspector, or plans examiner regardless of employer; a municipality recognizes and accepts a licensed firm's activities as its own and bears no liability for them.

*Source: OAR 918-098-1470; ORS 455.080; ORS 455.148(9) and (10).*

Stop-work authority is local, not a residential-code provision.

The ORSC does not adopt the model code's stop-work-order section (R114); stop-work authority comes from municipal ordinance, which the state building code statute preserves.

*Source: ORS 455.020(4).*

## 2.7 Owner-Builder Position In The Code Map

The owner-builder path is a legal role with specific duties. It decides who pulls each permit, who signs the homeowner statement, who may perform regulated work, who hires licensed subcontractors, who schedules inspections, and who answers for the jobsite.

There is no statewide three-year resale clock for owner-built homes. Oregon uses several separate rules instead:

- an intent-to-sell test in contractor-licensing law (ORS 701.010(5))
- a five-year cooldown on the structural-code exemption (ORS 455.320)
- a separate five-year cooldown on the rural-county exemption (ORS 455.335)
- a 75-day post-completion lien-notice window (ORS 87.007 and 87.093)
- the federal capital-gains rules (IRC section 121)

The legal chapter lists each one with its citations and resolved position.

Before regulated work begins, the roles are assigned. Homeowner, owner-builder, contractor, trade contractor, engineer, inspector, jurisdiction, and brand entity are not interchangeable. The permit, the license, the inspection record, and the certificate of occupancy each name someone, and the names must match the actual work.

Four trade-side rules are hard boundaries:

- LP gas: no homeowner exemption. Any LP gas installation, fitting, or piping work requires a State Fire Marshal licensed installer (ORS 480.432).
- Septic: a separate DEQ permit path. The owner may install under their own permit (OAR 340-071).
- Water wells: a separate OWRD path. The owner may drill under a Landowner Well Permit (OAR 690-205-0175).
- Refrigerant: federal. Connecting, gauging, or charging refrigerant lines requires an EPA Section 608 certified technician, regardless of homeowner status (40 CFR 82).

## 2.8 Inspection Checkpoints

Inspections are scheduled gates in the build sequence. Each one verifies a stage before the work gets covered, and each one creates a record.

**Simple**

The build schedule is organized around the inspections. Concrete cannot be poured until the footing inspection passes. Walls cannot close until the rough plumbing, electrical, and mechanical inspections pass. Nobody moves in until the final inspections pass. A missed inspection means uncovering or demolishing finished work so the inspector can see what is underneath.

**Applied**

The major checkpoints for a concrete house:

| Inspection | When | What gets verified |
|---|---|---|
| Permit application and plan review | Before any regulated work | Approved drawings, calculations, engineering |
| Erosion control or site inspection | Where the jurisdiction requires it | Erosion measures, site-disturbance limits |

| | | |
|---|---|---|
| Footing | Before concrete placement | Footing geometry, reinforcement, anchor bolts, bearing surface, UFER ground |
| Foundation drainage and dampproofing | Before backfill | Drain rock, perimeter drain, membrane, capillary break |
| Under-slab | Before slab placement | Subgrade, capillary break, vapor retarder, soil-gas retarder where required, insulation, sleeves |
| Rough plumbing | Before walls or chases close | DWV test, water-supply rough, gas pressure test where applicable |
| Rough electrical | Before walls close | Boxes, conductors, panel rough, grounding, bonding, low voltage where included |
| Rough mechanical | Before walls or ceilings close | Equipment, ducts, gas piping, ventilation, exhaust, refrigerant lines |
| Insulation and air sealing | Before drywall | Insulation type, depth, continuity, sealed penetrations |
| Energy and blower door | Where the compliance path requires testing | Blower-door result, duct-leakage result if applicable |
| Final building, electrical, plumbing, mechanical | Each discipline separately | Each discipline's final record |
| Site-development final | Before CO | Permanent erosion control, stormwater, address |
| Certificate of occupancy | All disciplines final, Form 4823 submitted | CO issued; binder closes |

> The Inspection And Approval Sequence chapter lists the readiness criteria for each gate, the corrective steps when work fails, and the records that result.

> For concrete work, schedule the inspection as part of the pour: trucks arrive in a fixed window, weather changes, forms can move, and curing starts whether the paperwork is ready or not. An inspector cannot approve work that is already hidden, so the inspection comes before the pour.

## 2.9 Code Blocks In This Book

Technical chapters state the rules in `Code` blocks. Every entry has three parts: what must happen, in plain words; the exact limit, value, or trigger; and the source. The source is the book name and section number, with the edition when it matters. An entry with no source does not appear.

Where the answer depends on the jurisdiction, the code edition, the site, or the product listing, the entry states the dependency and identifies what controls the local answer.

These entries let most decisions proceed without opening the official code text. The code lookup appendix and the glossary at the back provide the source location for the cases that need the original.

## 2.10 The Project Binder

The binder contains the proof that the code path was followed, organized from permit application onward.

What goes in:

- permit applications, approvals, and revisions
- approved plans, including any sheets revised after issuance
- plan-review comments and the response to each
- inspection cards, approvals, and correction notices for each discipline
- the signed BCD Property Owner Statement and any local equivalent
- engineering letters, stamped sheets, calculations, and any inspection or materials-testing records the engineered design requires (concrete strength tests, reinforcement-placement verification)
- product data, listings, and manufacturer installation instructions
- concrete batch tickets, soil reports, septic approvals, well records, the OWRD landowner permit and well log, and the SFM tank report if one applies
- photographs taken before each cover stage
- subcontractor CCB and BCD license records and insurance certificates
- BCD Form 4823 contractor information notification
- the final approval and occupancy records (CO or TCO and any extensions)

The binder serves four functions: compliance proof during construction, the working reference at each inspection visit, documentation for repairs and resale, and the record from which a future occupant can understand the house without guessing. The Documentation, Maintenance, And Future Upgrades chapter specifies organization, retention, and digital backup.